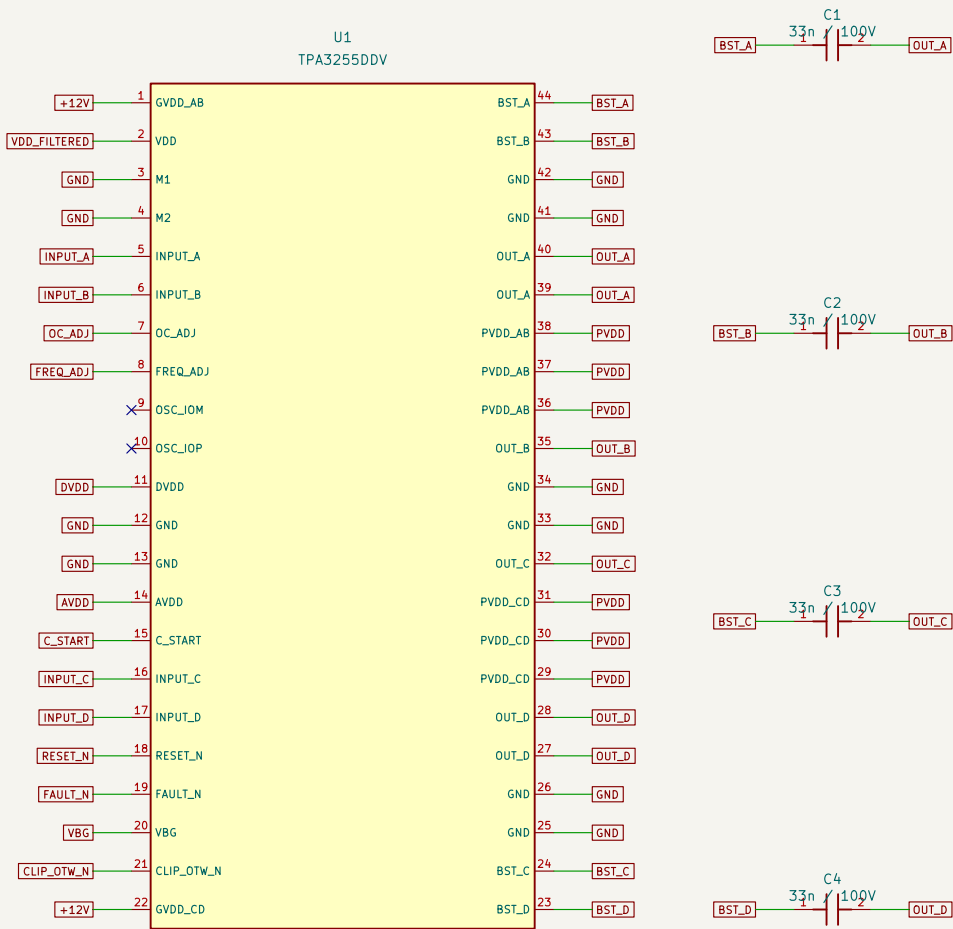


POWER STAGE / STEREO BTL / BENCH PROTOTYPE

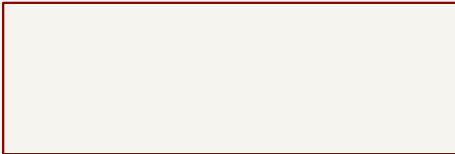
Global net names connect all eight sheets. Same labels are electrically connected.



M1=M2=0; OSC pins intentionally NC.

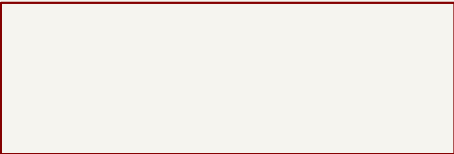
Top PowerPAD: ground heatsink per TI pin table.

control-supply



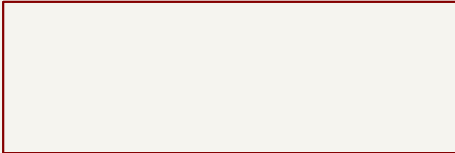
File: control-supply.kicad_sch

xlr-input



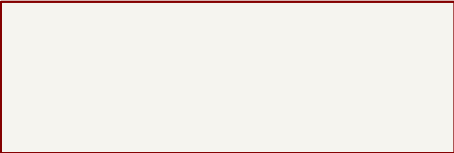
File: xlr-input.kicad_sch

input



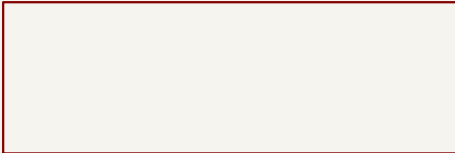
File: input.kicad_sch

power-control



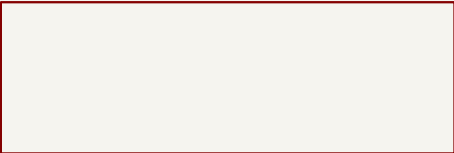
File: power-control.kicad_sch

feedback

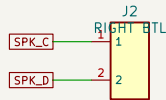
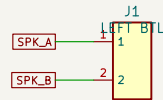


File: feedback.kicad_sch

trigger



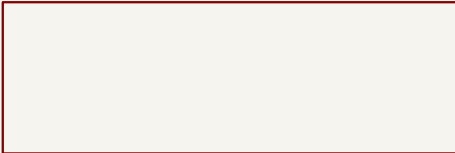
File: trigger.kicad_sch



Speaker terminals float: never join B or D to GND.

PFFB + LC: hardware loop stability still needs testing.

dc-switch



File: dc-switch.kicad_sch

Electrical draft – no PCB / footprints / hardware validation

Sheet: /
File: tpa3255-v03.kicad_sch

Title: TPA3255 V0.3 / tpa3255-v03

Size: A3 Date: 2026-09-11

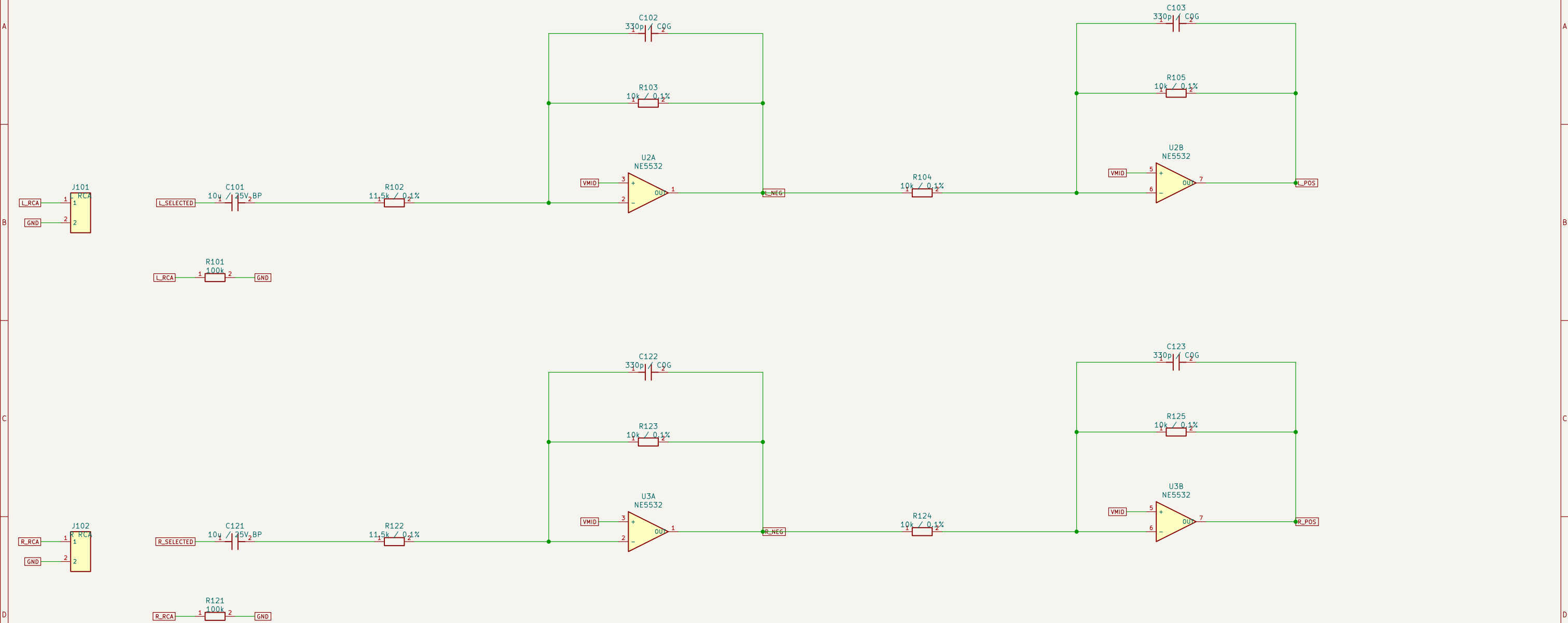
Rev: 0.3 REVIEW

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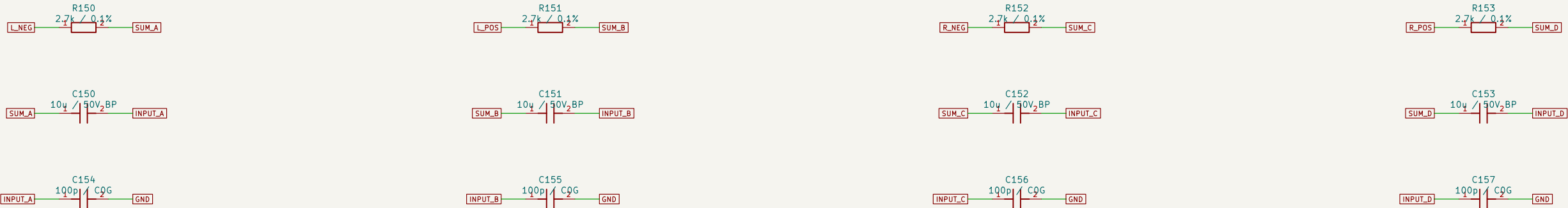
Id: 1/8

RCA / XLR SELECTED SIGNAL TO DIFFERENTIAL DRIVER

NE5532 on 12V, VMID approx. 6V. Signal capacitors are bipolar/film.



PFFB summing resistors BEFORE AC coupling; Z10 RCA 2.5Vrms target



Electrical draft – no PCB / footprints / hardware validation

Sheet: /input/
File: input.kicad_sch

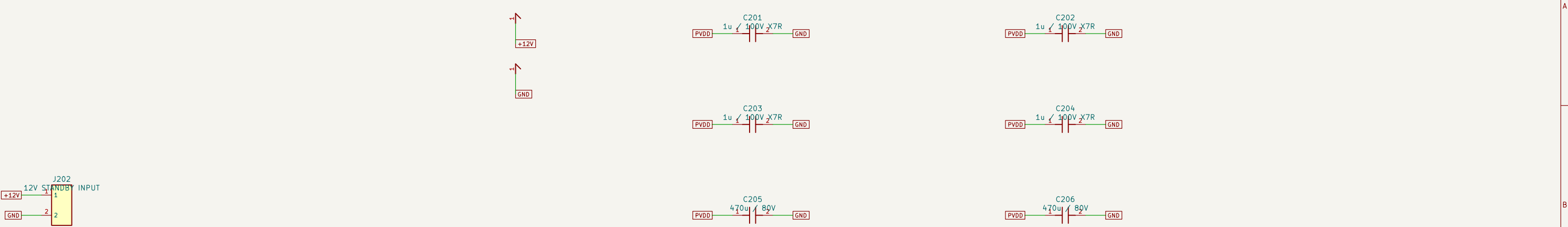
Title: TPA3255 V0.3 / input

Size: A3 Date: 2026-09-11
KiCad E.D.A. 10.0.6

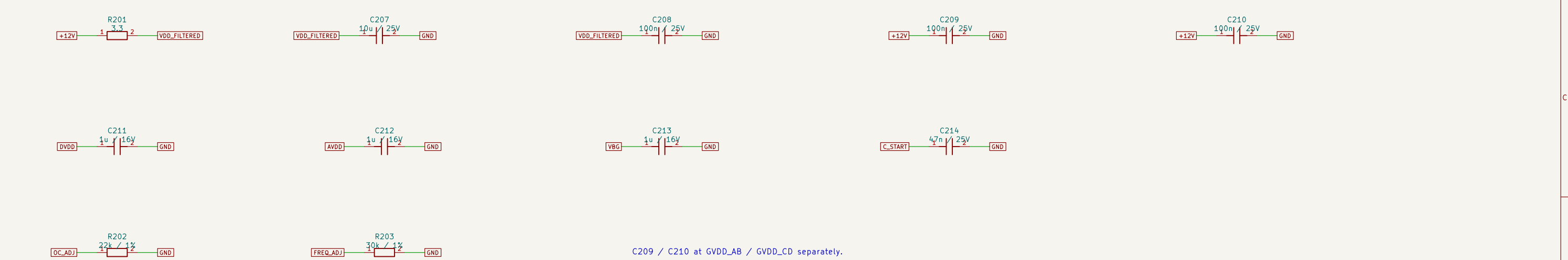
Rev: 0.3 REVIEW
Id: 2/8

12V STANDBY / LOCAL DECOUPLING

PVDD comes from dc-switch sheet. 12V standby remains on; no mains wiring on this PCB.



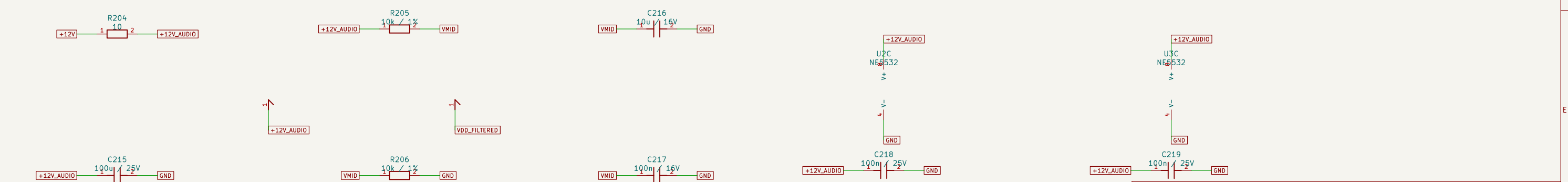
Place 2x MLCC + 1x bulk at EACH PVDD group.



C209 / C210 at GVDD_AB / GVDD_CD separately.

DVDD / AVDD / VBG: internal outputs, bypass only.

INPUT SUPPLY / VMID



Electrical draft – no PCB / footprints / hardware validation

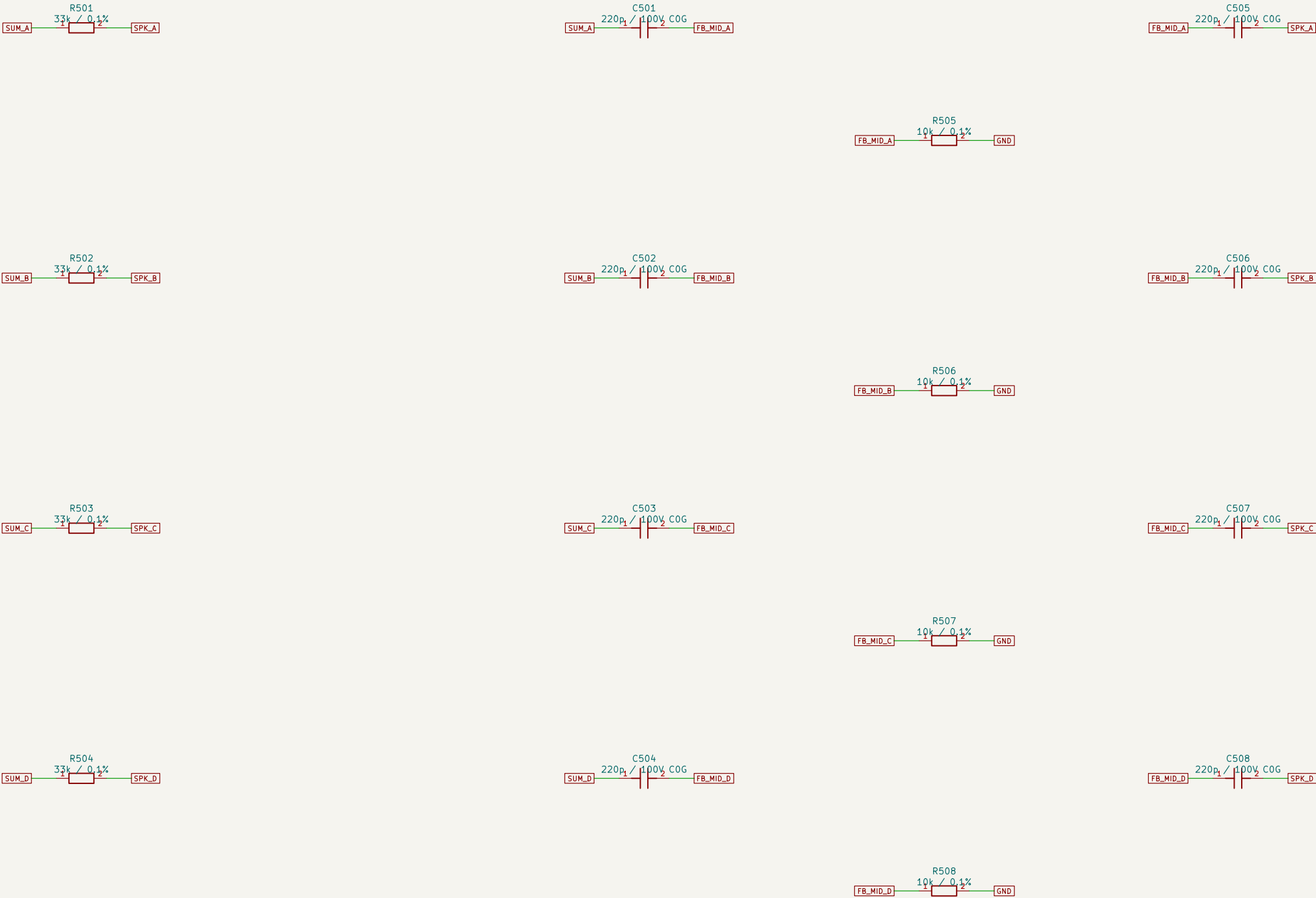
Sheet: /power-control/
File: power-control.kicad_sch

Title: TPA3255 V0.3 / power-control

Size: A3	Date: 2026-09-11	Rev: 0.3 REVIEW
KiCad E.D.A. 10.0.6		Id: 3/8

POST-FILTER FEEDBACK / TI SLAA788A TPA3255 STARTING VALUES

Same-arm negative feedback: SPK_A to SUM_A, etc. Summing node is BEFORE the input coupling capacitor.



PFFB, input bandwidth and LC are one loop: reduced model is NOT hardware stability proof.

Fit all four feedback branches; do not substitute R/C values without a loop review.

Electrical draft – no PCB / footprints / hardware validation

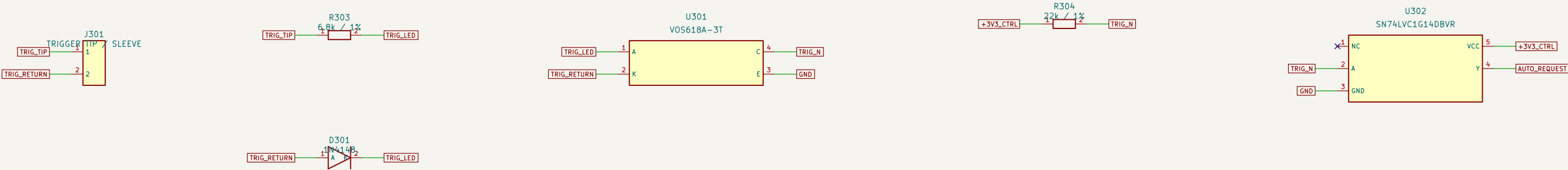
Sheet: /feedback/
File: feedback.kicad_sch

Title: TPA3255 V0.3 / feedback

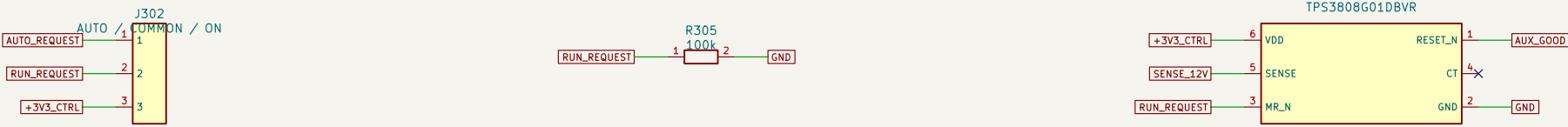
Size: A3	Date: 2026-09-11	Rev: 0.3 REVIEW
KiCad E.D.A. 10.0.6		Id: 4/8

ISOLATED TRIGGER / AUXILIARY MONITOR / POWER-OFF HOLD

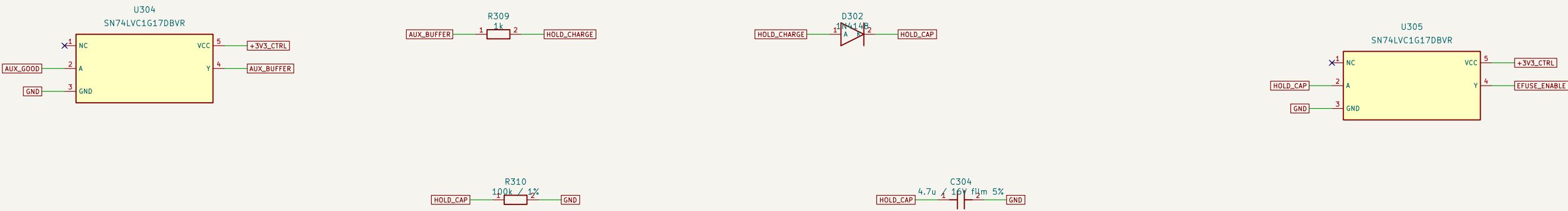
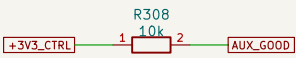
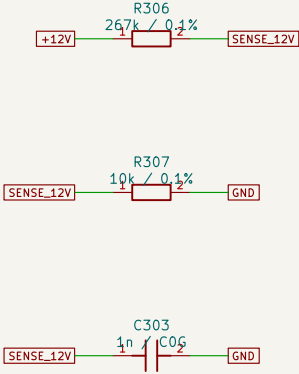
Design input: 9–15V DC, nominal 12V. Verify Z10 voltage, polarity and drive capability before connecting.



TRIG_RETURN is isolated from GND and chassis. Use an insulated jack; no trigger-powered relay coil.



External SPDT centre-OFF switch.



AUX_GOOD falls first to mute. HOLD_CAP keeps main DC on briefly; verify timing on hardware.

Electrical draft – no PCB / footprints / hardware validation

Sheet: /trigger/
File: trigger.kicad_sch

Title: TPA3255 V0.3 / trigger

Size: A3	Date: 2026-09-11	Rev: 0.3 REVIEW
KiCad E.D.A. 10.0.6		Id: 5/8

SWITCHED 48V DC / SLEW LIMIT / POWER-GOOD / AUDIO RESET

External regulated 48V source stays powered. Trigger-off disconnects PVDD; 12V standby remains on.

MODE open: latch fault; cycle Trigger to retry.

No external reverse-block FET. Surge testing pending.

PVDD > 40.9V, then ~1.03s delay.

Electrical draft – no PCB / footprints / hardware validation

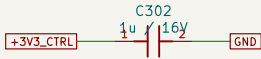
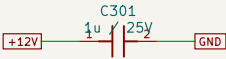
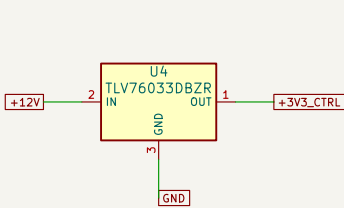
Sheet: /dc-switch/
File: dc-switch.kicad_sch

Title: TPA3255 V0.3 / dc-switch

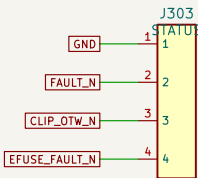
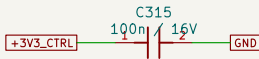
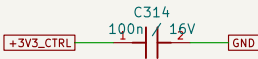
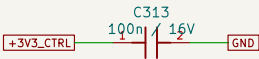
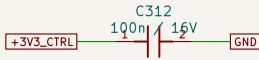
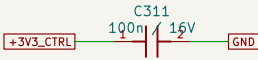
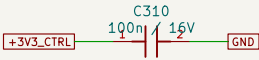
Size: A3	Date: 2026-09-11	Rev: 0.3 REVIEW
KiCad E.D.A. 10.0.6		Id: 6/8

Sheet: /dc-switch/		
File: dc-switch.kicad_sch		
Title: TPA3255 V0.3 / dc-switch		
Size: A3	Date: 2026-09-11	Rev: 0.3 REVIEW
KiCad E.D.A. 10.0.6		Id: 6/8

STANDBY CONTROL SUPPLY / LOCAL DECOUPLING / STATUS



C310–C315: one at each U302–U307 VCC/VDD pin on PCB.



High-impedance monitoring only; no external 5V pullups.

TPA fault protection remains internal; no automatic fault retry loop.

Trigger OFF enters standby, not AC mains isolation.

Electrical draft – no PCB / footprints / hardware validation

Sheet: /control-supply/
File: control-supply.kicad_sch

Title: TPA3255 V0.3 / control-supply

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-11

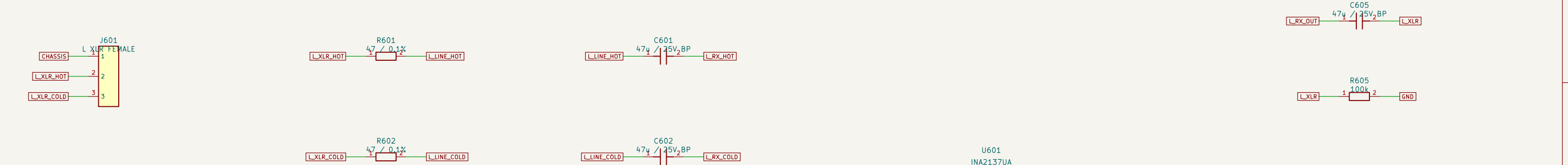
Rev: 0.3 REVIEW

Id: 7/8

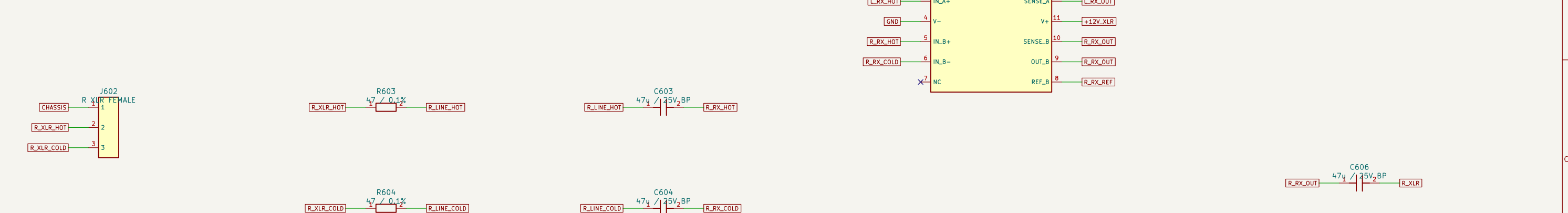
TRUE BALANCED XLR / INA2137 GAIN 0.5 / RCA SELECTION

XLR female: 1 shield, 2 hot, 3 cold. Both signal legs are AC coupled; do not ground pin 3.

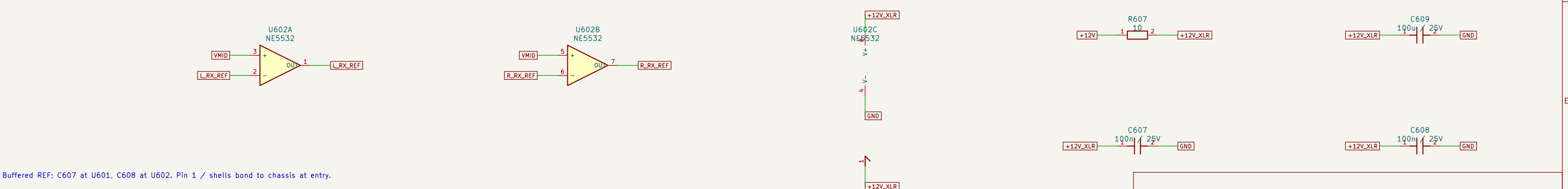
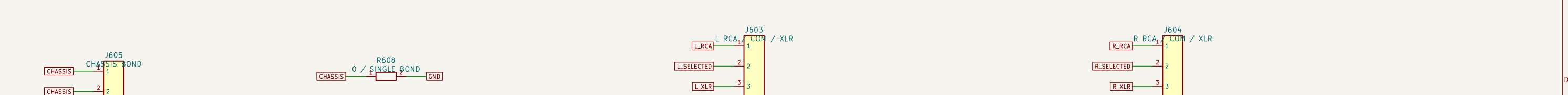
A



C601/C602 and C603/C604: match pairs to 1% measured capacitance.



J603/J604: fit 2–3 for XLR, 1–2 for RCA. Change both only in standby.



Buffered REF; C607 at U601, C608 at U602. Pin 1 / shells bond to chassis at entry.

Internal balanced-to-single-ended receiver; this is not a fully differential end-to-end amplifier.

Electrical draft – no PCB / footprints / hardware validation

Sheet: /xlr-input/
File: xlr-input.kicad_sch

Title: TPA3255 V0.3 / xlr-input

Size: A3 Date: 2026-09-11

Rev: 0.3 REVIEW

KiCad E.D.A. 10.0.6

Id: 8/8